

TrueYield recovery report — Andre Agassi Preparatory Academy - Building A (system 34)

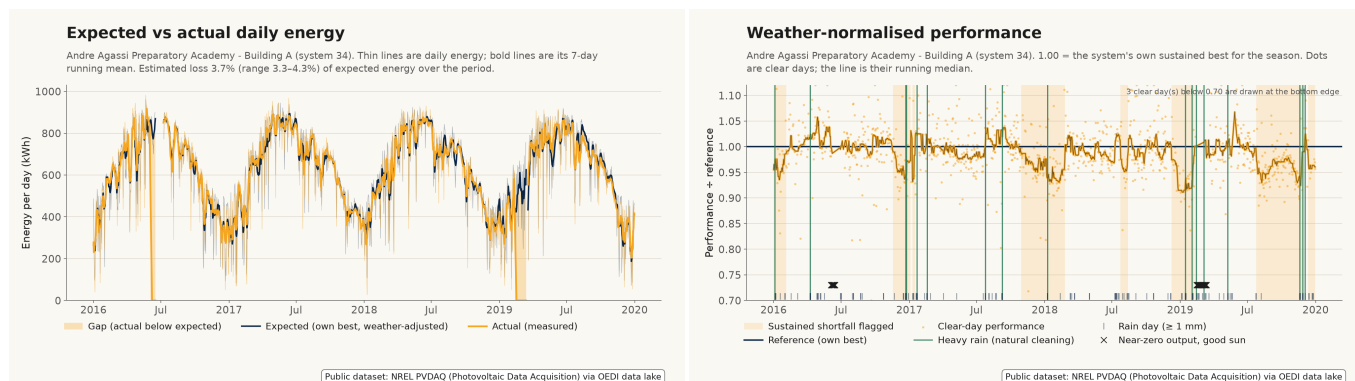
Data label: Public dataset: NREL PVDAQ (Photovoltaic Data Acquisition) via OEDI data lake | Period 2016-01-01 to 2019-12-31 | 146.64 kWp, tilt 11.2°, azimuth 180° | generated 2026-10-04

1. What is measured (facts)

- Generation: 1430 valid days of 1461 (97.9% complete); 25 days with no data, 6 partial days and 0 days with impossible values were excluded, not filled in.
- Loader inferred: pvdaq_system_34.csv: system 34; time column 'measured_on'; power from ['ac_power_hw_2695'] in mixed units; 15-minute steps. unit inferred from peak power vs 146.64 kWp nameplate, not from the column name. the unit of this column changes over time: hW (x100 W) until 2018-08-03, W from 2018-08-04. energy = power x sampling interval; missing samples stay missing.
- Weather: NASA POWER satellite irradiance (all-sky and clear-sky), 2 m temperature and rainfall for the site coordinates. No on-site sensor is used.

2. What is estimated, and how sure

- Estimated loss: 3.7% of expected energy (range 3.3–4.3%), 32,978 kWh (range 29,363–38,033 kWh) over 1430 valid days. Of this, 9,305 kWh (range 5,726–14,022) is sustained shortfall in 9 flagged episode(s), and 23,673 kWh (range 23,283–24,228) is from days with near-zero output. Ranges are 95% bootstrap intervals covering day-to-day noise and the uncertainty of the reference. Shortfalls below the trigger are counted as zero, so this is a conservative figure.
- Detection limit: losses below about 4.0% sustained over 14 days, or about 3.9% sustained over 30 days, cannot be distinguished from noise (from clear-day scatter of 3.5% per day; 1% false-alarm rate per window, 80% power, starting from a system at its reference). Checked with injected-loss trials on real data from system 1420: reliable detection (at least 80% of trials) was reached at about 4% over 14 days, and at about 4% over 30 days. Smaller losses are counted as zero.
- Method: Hourly mode. NASA POWER hourly global irradiance is split into beam and diffuse (Erbs), transposed to the array plane (Hay-Davies, pvlib) and converted to DC energy with a PVWatts model (temperature coefficient -0.4%/°C, NOCT 45 °C). Hourly values are summed to days. Performance index = measured ÷ modelled energy. The reference is the system's own 90th-percentile sustained (7-clear-day median) performance per calendar month, pooled over all years and reduced by the upward bias expected from noise. Clear day = all-sky ÷ clear-sky irradiation ≥ 0.85. Assumptions: constant inverter and wiring losses are absorbed by the reference; timestamps are local standard time; the cleanest state seen in each month is representative of a clean array.



3. Likely causes: detected facts vs hypotheses

Cause	Detected facts	Hypothesis (not proven)
outage or fault indicated	Output was near zero on 37 day(s) with good irradiance: 2016-06-04 to 2016-06-15 (12 days); 2019-02-15 to 2019-03-05 (19 days); 2019-03-07 to 2019-03-09 (3 days); 2019-03-11 to 2019-03-13 (3 days). 25 day(s) have no usable generation data (nothing logged, or a reading missing from at least one inverter); whether the system produced on those days is unknown and they are not counted as loss.	Near-zero days are consistent with an inverter trip, grid outage or a monitoring fault that recorded zeros. The data cannot tell these apart. Near-freezing temperatures on 1 of these days mean snow cover is also possible.
soiling not indicated	Between cleaning events, performance declined in 24 of 24 usable intervals, at a median of 0.159% per day in the declining ones. 1 of 14 testable natural rain event(s) were followed by a rise in normalised performance that is distinguishable from zero. rdttools estimates a soiling loss of 4.5% (range 2.7–7.0%, rdttools 3.2.1 stochastic rate and recovery (SRR)). Shuffle check: with the same clear-day values in random time order, which contain no soiling pattern, it reports 3.8–5.2%.	No soiling distinguishable from scatter. The estimate of 4.5% (range 2.7–7.0%, rdttools 3.2.1 stochastic rate and recovery (SRR)) is not clearly above what the same method reports on shuffled data (up to 5.2%), so most or all of it can be produced by day-to-day scatter alone.
ageing or degradation indicated	Year-on-year change of clear-day performance over 4.0 years: -0.95% per year (range -1.58 to -0.42% per year, rdttools 3.2.1 year-on-year).	Pattern consistent with gradual ageing or another slow loss not reversed by rain: the whole range is below zero.

shading indicated	Clock check: on clear days the measured daily profile is centred -3 minutes from the modelled one (-2 min in May-August, -4 min in November-February). 14:00-15:00: output was a median 18% below the midday level on most clear days in month(s) [12]. 15:00-16:00: output was a median 24% below the midday level on most clear days in month(s) [1]. 16:00-17:00: output was a median 23% below the midday level on most clear days in month(s) [3]. 17:00-18:00: output was a median 28% below the midday level on most clear days in month(s) [6]. Symmetry check: with the sun 15-25° up, output is at 99% of the midday level in the morning and 88% in the afternoon.	Pattern consistent with shading or a horizon obstruction in the afternoon. A modelling error at low sun would lower morning and afternoon alike; here the afternoon is clearly lower at the same sun height, which points to the site: an obstruction on that side, or an array facing slightly away from the stated azimuth. It needs a site check.
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4. Recovery verification

- Events tested: natural rain event, not a maintenance intervention. 14 of 17 event(s) had enough clear days on both sides (14 days before, 14 after).
- Best-observed event, 2017-07-25 (natural rain event, not a maintenance intervention): +3.1% (range -2.4 to +7.9%), +21.9 kWh/day (range -16.9 to +54.9), from 9 clear days before and 10 after. It was chosen for having the most clear days, not for its size. The range includes zero: no recovery can be claimed for this event.
- Across all testable events: 1 clear rise(s), 0 clear fall(s), 13 not distinguishable from zero. Full list in summary.json.

5. Value and verdict

- [not available: tariff_inr_per_kwh not provided in data/systems.csv]
- Emissions: [not available: emission_factor_kg_per_kwh is empty in config.yaml; no factor is assumed]

6. Limits of this report

- Causes are hypotheses; none is proven without a site inspection.
- Loss is measured against this system's own best month by month; a loss present from the start, or one that recurs in the same month every year, is partly or wholly invisible.
- Irradiance is a satellite area average; losses below the detection limit are not counted.

7. Method validation with injected loss — not a field result

- On a clean real stretch of system 1420 (2018-04-13 to 2018-12-08), a 10% soiling ramp and a 3-day outage were injected (1,221 kWh). The unchanged pipeline recovered 1,041 kWh (85% of the injected loss; estimate 1,041 kWh, range 762-1,610); the known truth is inside the stated range. Outage days found: 3 of 3. Detection-limit check: over 14 days: 2% step found in 27%, 4% step found in 80%, 6% step found in 93%, 8% step found in 97%, 10% step found in 100%, 12% step found in 100%; over 30 days: 2% step found in 47%, 4% step found in 90%, 6% step found in 100%, 8% step found in 100%, 10% step found in 100%, 12% step found in 100% of 30 trials each. These are test results on altered data, not field results.